LEARNER GUIDE

Expanding three brackets

Algebraic Expressions and Identities · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To expand three brackets, expand two of them first, then multiply the result by the third. Every product takes one term from each bracket, so there are 2 × 2 × 2 = 8 products before collecting. Powers of x add when you multiply: $x \times x^2 = x^3$. For $(x + a)(x + b)(x + c)$, the number term is abc and the x^2 coefficient is $a + b + c$: two quick checks.

What you are learning to notice

- Stage 1 is a 2 by 2 grid. Stage 2 puts the stage 1 answer down the side of a 3 by 2 grid.
- In stage 2, like terms sit on the diagonals of the grid, so collecting is quick.
- A box with edges $(x + 1)$, $(x + 2)$ and $(x + 3)$ splits into 8 pieces: one x^3 cube, three slabs, three rods and one small block.
- $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$.
- If a bracket starts with $2x$, the x^3 coefficient is 2, and the quick checks change.

Moving through it, one step at a time

Choose two brackets and expand them with a 2 by 2 grid; collect. Put that answer down the side of a new grid, with the third bracket as the column headings. Fill every cell, collect like terms, and check: the number term should be the product of the three numbers.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Two stages

Expand and simplify $(x + 1)(x + 2)(x + 3)$.

Stage 1: $(x + 1)(x + 2)$

×	x	**2**
x	x^2	$2x$
1	x	2

Stage 2: $(x^2 + 3x + 2)(x + 3)$

×	x	**3**
x^2	x^3	$3x^2$
3x	$3x^2$	$9x$
2	$2x$	6

1. Stage 1: fill the 2 by 2 grid, then collect.	$x^2 + 2x + x + 2 = x^2 + 3x + 2$
2. Stage 2: $x^2 + 3x + 2$ goes down the side; $(x + 3)$ gives the column headings. Fill every cell.	six cells
3. Collect: like terms sit on the diagonals (the shaded x^2 cells).	$x^3 + 6x^2 + 11x + 6$
4. Check: $1 \times 2 \times 3 = 6$, and $1 + 2 + 3 = 6$.	✓

$(x + 1)(x + 2)(x + 3) = x^3 + 6x^2 + 11x + 6$.

Example 2 — With negative numbers

Expand and simplify $(x – 2)(x + 3)(x + 1)$.

Stage 1: $(x – 2)(x + 3)$

×	x	**3**
x	x^2	$3x$
–2	$–2x$	–6

Stage 2: $(x^2 + x – 6)(x + 1)$

×	x	**1**
x^2	x^3	x^2
x	x^2	x
–6	$–6x$	–6

1. Stage 1: fill the grid, keeping each sign with its term, then collect.	$x^2 + 3x – 2x – 6 = x^2 + x – 6$
2. Stage 2: $x^2 + x – 6$ down the side, $(x + 1)$ as the column headings.	six cells
3. Collect along the diagonals.	$x^3 + 2x^2 – 5x – 6$
4. Check: $(–2) × 3 × 1 = –6$, and $–2 + 3 + 1 = 2$.	✓

$(x – 2)(x + 3)(x + 1) = x^3 + 2x^2 – 5x – 6$.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (expanding two brackets with the grid, and multiplying negative numbers), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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